UNIVERSITY OF ROME “TOR VERGATA”

Faculty of Engineering

M.Sc. in Management Engineering

COURSE

Management of Innovation and Technology

Autodesk Fusion 360

Project Report

Project 2 – Injection Molding Simulation

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1. Simulation Setup & Material Selection

All 18 simulations were conducted in Autodesk Fusion (Injection Molding Simulation module) using a single-solid target body representing a **Remote Control Housing**. The geometry features varying wall thicknesses, circular button cutouts, internal bosses, and rounded edges — all of which are known to promote weld lines and sink marks in injection-moulded parts.

Primary material (Studies 1–17): Generic PP (Polypropylene, Crystalline). PP was chosen for its excellent flowability, balanced mechanical properties, and cost-effectiveness for consumer electronics enclosures. Its semi-crystalline structure makes it sensitive to mold temperature variations, which is clearly reflected in the shrinkage and warpage results across the study set.

Comparison material (Study 18): Generic ABS (Acrylonitrile Copolymers, Amorphous). Study 18 reused the best PP configuration (Study 9) with the material changed to ABS to evaluate the impact of switching from a crystalline to an amorphous polymer. The ABS comparison is detailed in Section 7.

Parameter	PP Studies (1–17)	ABS Study (18)
Material	Generic PP	Generic ABS
Structure	Crystalline	Amorphous
Baseline Melt Temperature	493.15 K (220 °C)	493.15 K (220 °C)
Baseline Mold Temperature	323.15 K (50 °C)	303.15 K (30 °C)
Injection Time	Automatic (all)	Automatic
Aesthetic Faces	146 (from Study 4)	146
Total Studies	17 (incl. 2 repeats)	1

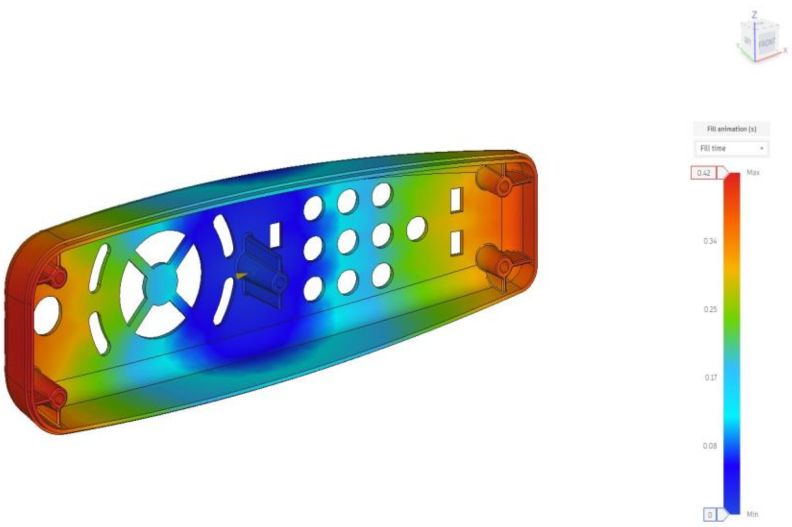
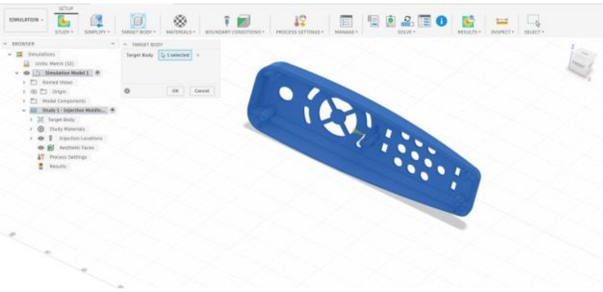


Figure 1 — Fill animation, Study 1 (1 gate, original geometry). Blue = filled first; red = last.

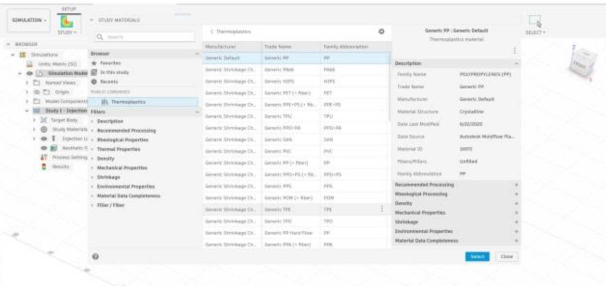
2. Simulation Methodology — Steps Performed for Each Test

Each of the 18 simulation studies followed the same eight-step procedure in Autodesk Fusion. The screenshots below document these steps as performed for a representative study. Parameters (gate count, temperatures) were changed between studies but the setup workflow remained identical throughout.



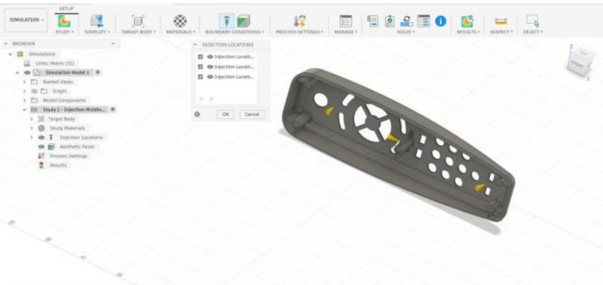
Step 1: Target Body Selection

The remote control housing body is selected as the single simulation target.



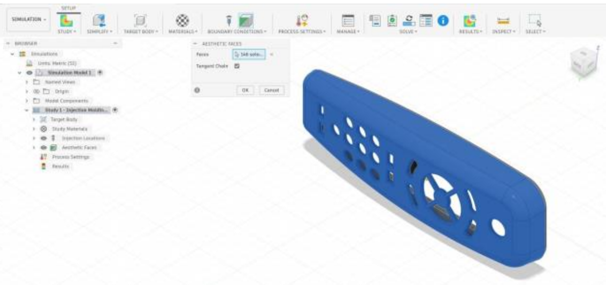
Step 2: Material Selection

Generic PP (Polypropylenes family, Crystalline) chosen from the Autodesk material library.



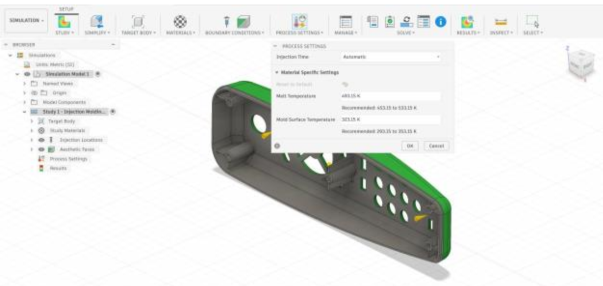
Step 3: Injection Location Placement

Gate point(s) positioned on the non-aesthetic (bottom) face of the housing.



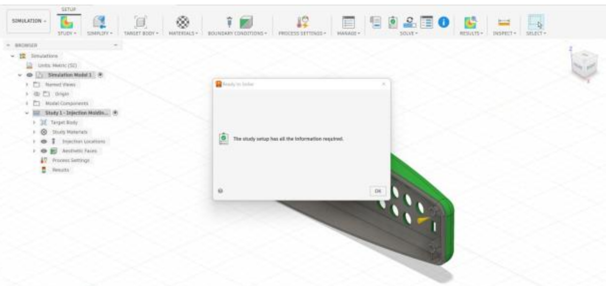
Step 4: Aesthetic Faces Designation

146 top-face surfaces selected as aesthetic faces requiring highest surface quality.



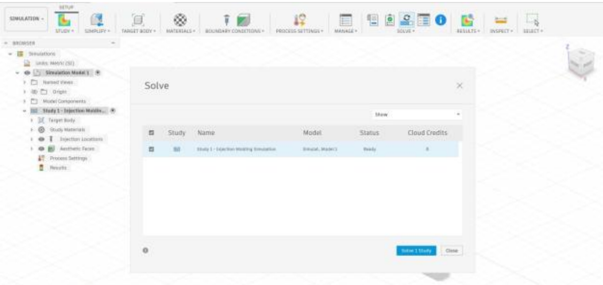
Step 5: Process Settings

Melt temperature, mold surface temperature, and injection time (Automatic) configured.

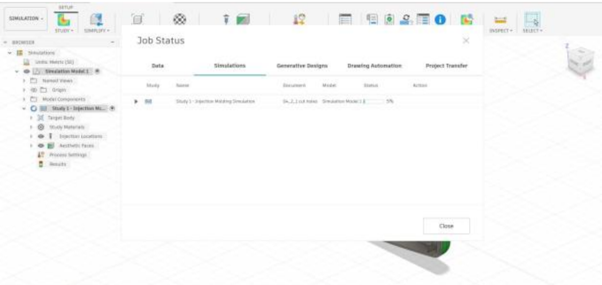


Step 6: Pre-Check — Ready to Solve

Fusion confirms all required inputs are present before cloud submission.



Step 7: Solve Dialog
Study submitted to Autodesk cloud solver. Each run consumes ~6 cloud credits.



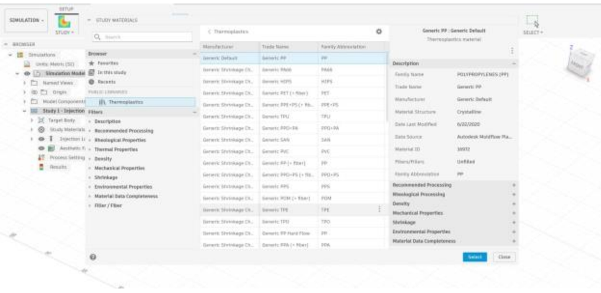
Step 8: Job Status Monitor
Cloud solve progress tracked; results downloaded automatically on completion.

Step 1 — Target Body: The remote control housing solid body is selected as the injection-moulding simulation target. Only one body was simulated per study. **Step 2 — Material:** The material library was opened and the appropriate thermoplastic selected (Generic PP for Studies 1–17, Generic ABS for Study 18). **Step 3 — Injection Locations:** Gate points were placed on the non-aesthetic (bottom) face. The number and position of gates varied by study (1, 2, 3, or 4 gates). **Step 4 — Aesthetic Faces:** From Study 4 onwards, 146 top-face surfaces were designated as aesthetic faces, directing the solver to evaluate weld lines and sink marks on those surfaces with greater precision. **Step 5 — Process Settings:** Injection time was set to Automatic throughout; melt temperature and mold surface temperature were varied per study. **Step 6 — Pre-Check:** Fusion's built-in pre-check confirmed all required inputs were complete before submission. **Step 7 — Solve:** Each study was submitted to the Autodesk cloud solver (~6 cloud credits per run). **Step 8 — Job Status:** Solve progress was monitored via the Job Status panel; results were downloaded automatically on completion.

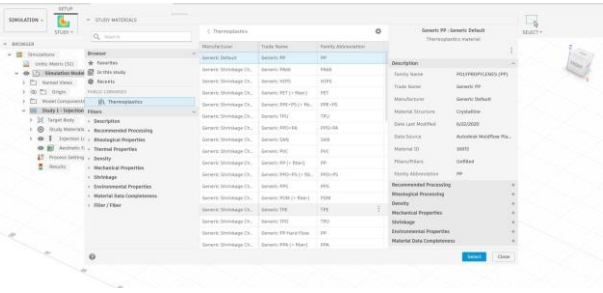
3. Geometry Modification (Studies 1–3 → Studies 4–18)

After the first three simulation studies on the original geometry, the remote control housing was modified to improve its suitability for injection moulding. The principal change was the simplification of the large internal boss and rib cluster located at the top end of the housing interior. In the original geometry, this feature comprised multiple tall, closely-spaced ribs creating a zone of significantly greater wall thickness — a known source of sink marks and flow-front disturbance. The modified geometry reduces the height and density of these internal features, producing more uniform wall thickness throughout the cavity.

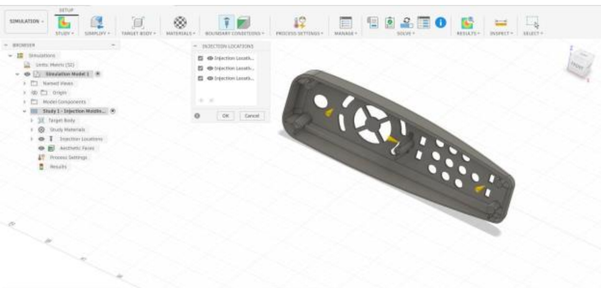
The geometry change was implemented in Autodesk Fusion's Design workspace. The eight steps below document the modification process, followed by before/after views of the component.



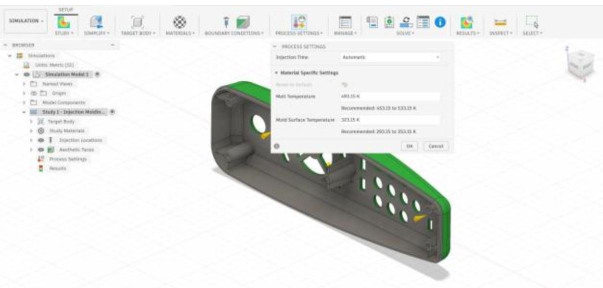
Geometry Step 1



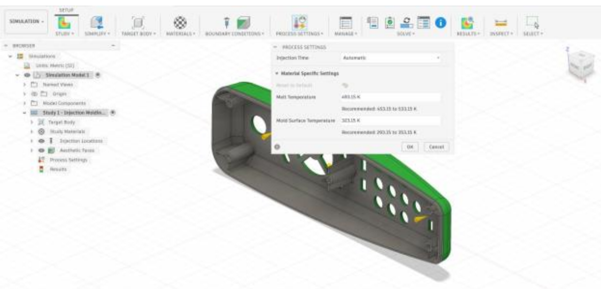
Geometry Step 2



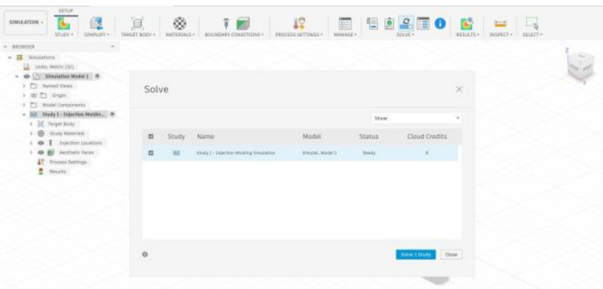
Geometry Step 3



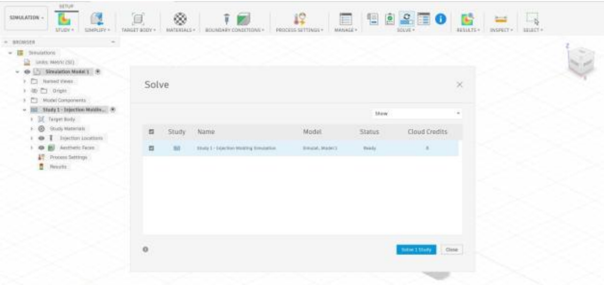
Geometry Step 4



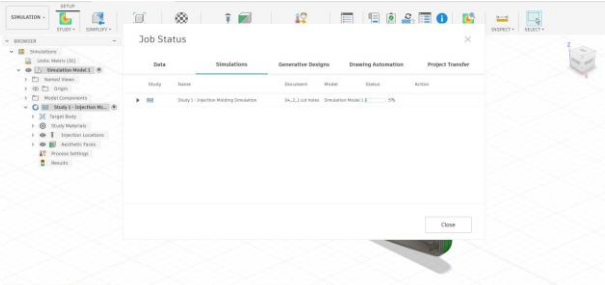
Geometry Step 5



Geometry Step 6

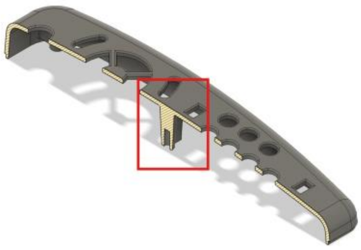


Geometry Step 7

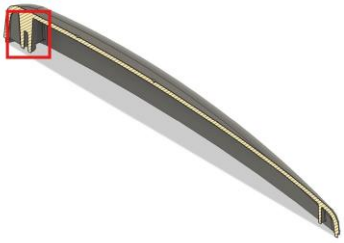


Geometry Step 8

Before the Modification (Studies 1–3 — Original Geometry)



Original geometry — view 1



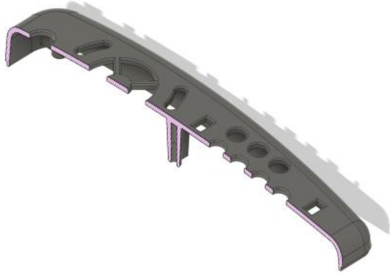
Original geometry — view 2

Original geometry — view 3

The original geometry features a prominent multi-rib boss cluster (highlighted by the red box in views 1 and 2) at the top end of the housing interior. The thick cross-section ribs create an area of substantially greater wall thickness than the surrounding shell, which the simulation software identifies as a primary sink-mark risk zone and a location prone to flow hesitation.

After the Modification (Studies 4–18 — Modified Geometry)

Modified geometry — view 1



Modified geometry — view 2

Modified geometry — view 3

The modified geometry retains all functional features of the remote control housing (button cutouts, perimeter walls, attachment bosses) but significantly reduces the height and density of the central internal rib cluster. This lowers the maximum local wall thickness, improves flow uniformity through the cavity, and reduces the thermal gradient that drives sink mark formation. The draft analysis overlays (pink hatching in after views) confirm adequate draft angles on all faces for clean ejection. The structural effect of this change is visible in the simulation results: from Study 3 (original geometry, 2 gates, 52 weld faces total) to Study 4 (modified geometry, 2 gates, 34 weld faces on 146 aesthetic faces).

4. Studies Overview

The table below summarises all 17 PP studies and Study 18 (ABS). Studies 1–3 used the **original geometry**; Studies 4–18 used the **modified geometry**. Defect face counts for Studies 1–3 span all model faces (aesthetic faces not yet designated); Studies 4–18 count on 146 aesthetic faces — these two populations are not directly comparable. Study 17 is a repeat of Study 9.

S#	Geom	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Shrink%	Weld	Sink	Mat
1	Orig	1	493.15K	323.15K	0.42	16.167	0.266	1.50	60*	—*	PP
2	Orig	1	493.15K	323.15K	0.42	16.167	0.266	1.50	60*	—*	PP
3	Orig	2	493.15K	323.15K	0.42	13.229	0.161	1.769	52*	—*	PP
4	Mod	2	493.15K	323.15K	0.42	13.229	0.161	1.769	34	7	PP
5	Mod	2	513.15K	323.15K	0.31	12.234	0.178	1.734	37	7	PP
6	Mod	2	533.00K	323.15K	0.21	11.987	0.212	1.701	34	7	PP
7	Mod	2	473.15K	323.15K	0.42	15.552	0.083	1.763	32	7	PP
8	Mod	2	493.15K	343.15K	0.42	12.761	0.254	1.776	37	7	PP
9	Mod	2	493.15K	303.15K	0.31	14.537	0.062	1.734	34	7	PP
10	Mod	4	493.15K	323.15K	0.31	6.652	0.831	1.87	42	5	PP
11	Mod	3	493.15K	323.15K	0.31	8.455	0.229	1.869	37	8	PP
12	Mod	1	493.15K	323.15K	0.42	16.189	0.269	1.509	35	7	PP
13	Mod	2	493.15K	343.15K	0.42	11.082	0.496	1.735	40	5	PP
14	Mod	3	493.15K	343.15K	0.31	6.65	0.443	2.035	39	8	PP
15	Mod	3	500.15K	343.15K	0.31	6.299	0.443	2.034	38	8	PP
16	Mod	3	505.15K	343.15K	0.31	6.096	0.464	2.051	41	8	PP
17	Mod	2	493.15K	303.15K	0.31	14.537	0.062	1.734	34	7	PP
18	Mod	2	493.15K	303.15K	0.63	42.972	0.123	0.437	34	5	ABS

* Studies 1–3: defect counts span all model faces (aesthetic faces not designated). Green = Study 9/17 (best deflection). Blue = Study 7 (fewest weld faces). Purple = Study 18 (ABS material).

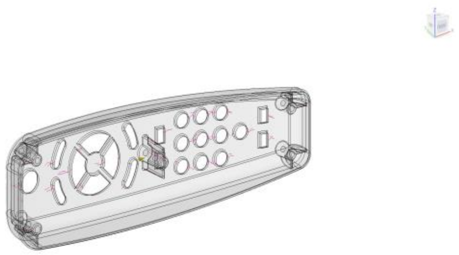
5. Phase-by-Phase Analysis

Phase 1 — Single-Gate Baseline (Studies 1, 2)

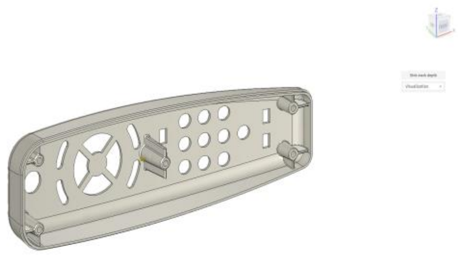
Original geometry. Single central gate. Baseline process parameters. No aesthetic-face designation.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
1	1	493.15K	323.15K	0.42	16.167	0.266	60	?
2	1	493.15K	323.15K	0.42	16.167	0.266	60	?

Studies 1 and 2 used the original geometry (before the modification described in Section 3) with a single central gate and standard PP baseline parameters (melt 493.15 K / mold 323.15 K, automatic injection time). The cavity filled in 0.42 s at 16.167 MPa. Defect counts here span all geometry faces because aesthetic faces had not yet been designated; 60 weld-line faces were reported across the full model.



Study 1: Weld lines



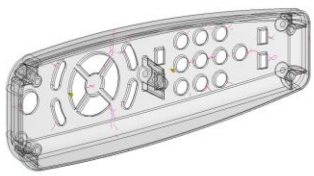
Study 1: Sink mark depth - Visualization

Phase 2 — Dual-Gate Baseline (Studies 3, 4)

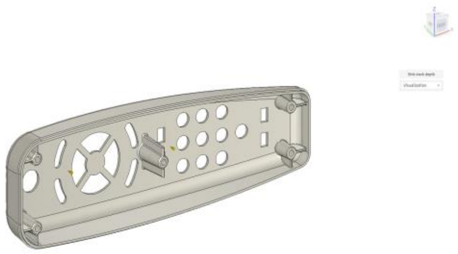
Study 3: original geometry, 2 gates. Study 4: modified geometry, 2 gates, 146 aesthetic faces activated.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
3	2	493.15K	323.15K	0.42	13.229	0.161	52	?
4	2	493.15K	323.15K	0.42	13.229	0.161	34	7

Study 3 kept the original geometry but added a second gate, reducing fill pressure by ~18 % (13.229 MPa) and total weld-line faces from 60 to 52. Study 4 introduced two critical changes simultaneously: (i) the geometry was modified as described in Section 3, simplifying the internal boss structure, and (ii) 146 aesthetic faces were designated for consistent defect counting. This combination reduced the weld-line count on aesthetic surfaces to 34, establishing the comparable baseline for all subsequent studies (4–17).



Study 4: Weld lines



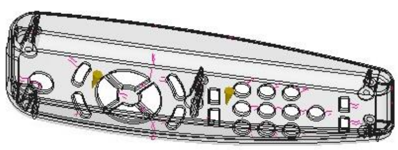
Study 4: Sink mark depth - Visualization

Phase 3 — Melt Temperature Sweep — 2 Gates (Studies 5, 6, 7)

Modified geometry. Two gates fixed. Melt temperature swept: 473 K ■ 493 K ■ 513 K ■ 533 K.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
5	2	513.15K	323.15K	0.31	12.234	0.178	37	7
6	2	533.00K	323.15K	0.21	11.987	0.212	34	7
7	2	473.15K	323.15K	0.42	15.552	0.083	32	7

With two gates and the modified geometry, melt temperature was swept around the baseline of 493.15 K. Study 5 raised it to 513.15 K; Study 6 to 533 K; Study 7 lowered it to 473.15 K. The cooler melt (Study 7) produced the fewest weld-line faces of the entire campaign (32) and a very low out-of-plane deflection (0.083 mm). Higher melt temperatures reduced pressure further but increased weld-line counts.



Study 7: Weld lines

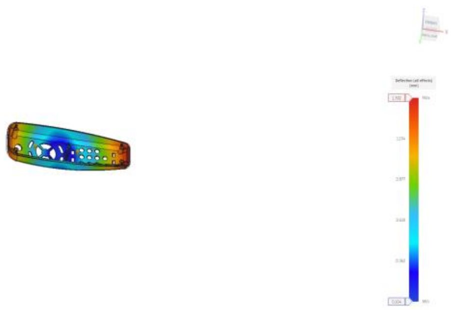
Study 7: Deflection (all effects)

Phase 4 — Mold Temperature Sweep — 2 Gates (Studies 8, 9)

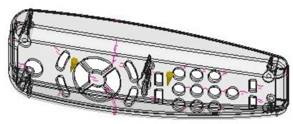
Modified geometry. Two gates. Mold temperature swept: 303.15 K (cold) vs 343.15 K (warm).

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
8	2	493.15K	343.15K	0.42	12.761	0.254	37	7
9	2	493.15K	303.15K	0.31	14.537	0.062	34	7

Mold surface temperature was varied while keeping two gates and standard melt (493.15 K). Study 8 used a warmer mold (343.15 K); Study 9 a colder mold (303.15 K). The cold-mold run (Study 9) achieved the single lowest out-of-plane deflection of all 17 PP studies — just 0.062 mm — while maintaining 34 weld-line faces on aesthetic surfaces and 7 sink-mark faces. This configuration was later selected as the best case and cloned for the ABS material comparison (Study 18). Study 17 is an independent verification repeat of Study 9 that confirmed all results identically.



Study 9: Deflection (all effects)



Study 9: Weld lines

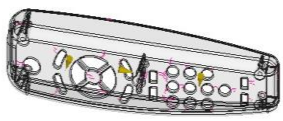
Phase 5 — Gate Count Exploration (Studies 10, 11)

Modified geometry. Gate count increased to 3 (Study 11) and 4 (Study 10); temperatures at baseline.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
10	4	493.15K	323.15K	0.31	6.652	0.831	42	5
11	3	493.15K	323.15K	0.31	8.455	0.229	37	8

Increasing gate count to three (Study 11) and four (Study 10) created more melt-front collision zones, paradoxically increasing weld-line counts. Four gates produced the highest weld count (42 faces) and worst deflection (0.831 mm) of the entire campaign, while three gates were only marginally better (37 weld faces, 0.229 mm). These results confirm that more gates do not guarantee better surface quality for this geometry.

Study 10: Deflection (all effects)



Study 11: Weld lines

Phase 6 — Verification & Alternate Config (Studies 12, 13)

Study 12: 1-gate aesthetic-face verification. Study 13: alternate 2-gate position with warm mold.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
12	1	493.15K	323.15K	0.42	16.189	0.269	35	7
13	2	493.15K	343.15K	0.42	11.082	0.496	40	5

Study 12 repeated the single-gate layout with aesthetic faces active, confirming the 1-gate baseline (35 weld faces, 0.269 mm). Study 13 used two gates with a warm mold (343.15 K) but at an alternate gate position, producing noticeably worse results (40 weld faces, 0.496 mm) than Study 8 with identical thermal settings — demonstrating the significant influence of gate placement on defect distribution.

Phase 7 — Three-Gate Thermal Optimization (Studies 14, 15, 16)

Modified geometry. Three gates + warm mold (343.15 K). Melt temperature sweep: 493 → 500 → 505 K.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
14	3	493.15K	343.15K	0.31	6.65	0.443	39	8
15	3	500.15K	343.15K	0.31	6.299	0.443	38	8
16	3	505.15K	343.15K	0.31	6.096	0.464	41	8

Three gates were combined with a warm mold (343.15 K) and a melt temperature sweep: 493.15 K (Study 14), 500.15 K (Study 15), 505.15 K (Study 16). All three delivered worse results than any two-gate optimised configuration: deflection ranged from 0.443 to 0.464 mm, weld faces from 38 to 41, and mold shrinkage reached its peak values (~2.0–2.05 %). The three-gate warm-mold combination consistently underperformed.

Study 14: Weld lines

Study 14: Deflection (all effects)

Phase 8 — Verification Repeat (Studies 17)

Independent repeat of Study 9 to verify cold-mold result. All metrics matched identically.

Study	Gates	Melt T	Mold T	Fill(s)	P(MPa)	Defl(mm)	Weld	Sink
17	2	493.15K	303.15K	0.31	14.537	0.062	34	7

Study 17 is an exact repeat of Study 9 (2 gates, melt 493.15 K, mold 303.15 K) run independently to verify the cold-mold result. Every metric matched identically: fill time 0.31 s, pressure 14.537 MPa, deflection 0.062 mm, shrinkage 1.734 %, 34 weld faces, 7 sink-mark faces. The perfect agreement confirms the simulation is stable and reproducible under these conditions.

6. Comparative Analysis

The following analysis covers Studies 4–17 (comparable scope — 146 aesthetic faces, modified geometry). Studies 1–3 used the original geometry and different defect-counting scope; they appear in charts for context but are excluded from the ranking.

6.1 Surface Defects: Weld Lines & Sink Marks

Weld lines form where two advancing melt fronts converge; their count on aesthetic faces is the primary surface-quality metric. Sink marks result from localised volumetric shrinkage beneath thick features. The face counts below are all on the 146 designated aesthetic faces (Studies 4–17).

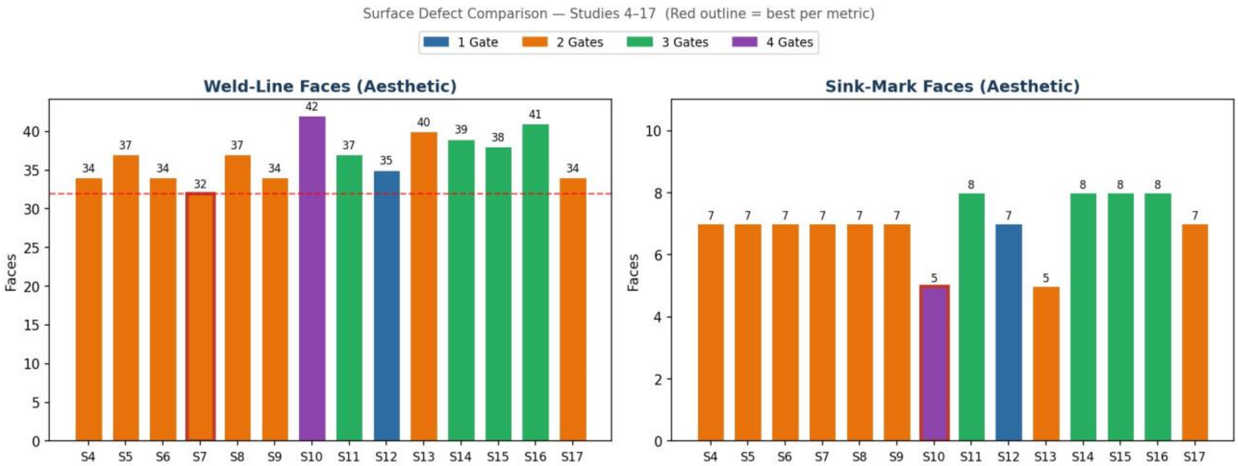


Figure 2 — Weld-line and sink-mark face counts for Studies 4–17. Red outline = minimum per metric. Bar colour = gate count.

Weld-Line Face Count — Top 8 (Studies 4–17):

Rank	Study	Gates	Melt T	Mold T	Weld Faces	vs Baseline
1	7	2	473.15K	323.15K	32	−2
2	4	2	493.15K	323.15K	34	baseline
3	6	2	533.00K	323.15K	34	baseline
4	9	2	493.15K	303.15K	34	baseline
5	17	2	493.15K	303.15K	34	baseline
6	12	1	493.15K	323.15K	35	+1
7	5	2	513.15K	323.15K	37	+3
8	8	2	493.15K	343.15K	37	+3

Study 7 achieves the minimum weld count (32 faces) by using a cooler melt (473.15 K, −20 K from baseline). Lower melt temperature increases polymer viscosity, slowing the flow front and reducing the severity of melt-front merging at features. Studies with more gates (10, 11) paradoxically increased weld counts by creating additional internal collision zones.

6.2 Dimensional Stability: Deflection & Shrinkage

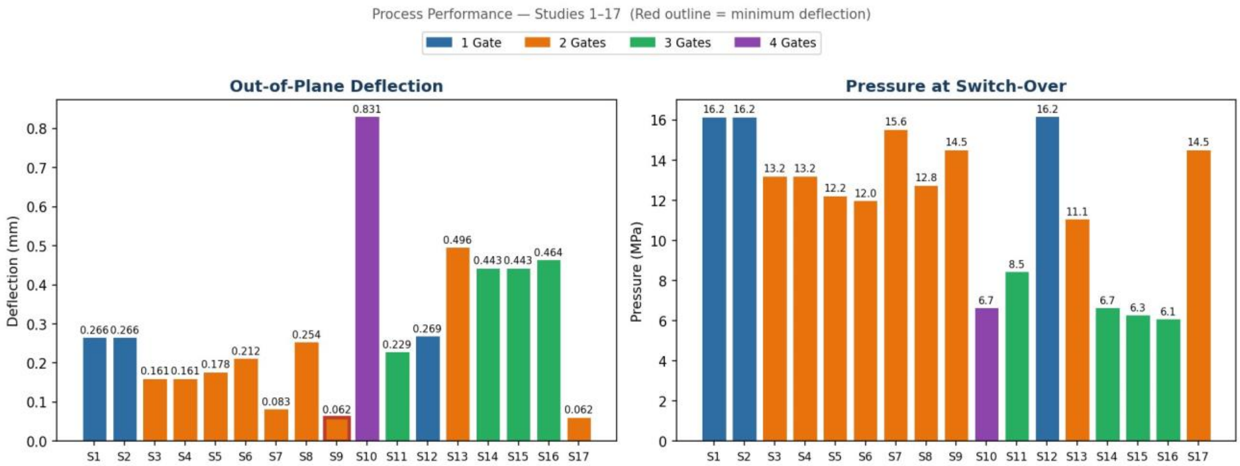


Figure 3 — Out-of-plane deflection and fill pressure for all 17 PP studies. Red outline = minimum deflection (Studies 9 and 17).

The dominant deflection driver is mold temperature. Study 9 (cold mold, 303.15 K) achieved 0.062 mm — a 77 % reduction versus the two-gate baseline (Study 4, 0.161 mm). Faster solidification in a cold mold reduces residual thermal gradients that cause warpage. Four gates (Study 10, 0.831 mm) produced the worst deflection due to asymmetric pressure distribution.

6.3 Top 5 Candidate Studies Comparison

From Studies 4–17, five configurations were selected as top candidates based on the combined balance of weld-line count, sink-mark count, and out-of-plane deflection. All five use two gates and the modified geometry.
The chart and table below compare them across all four key metrics.

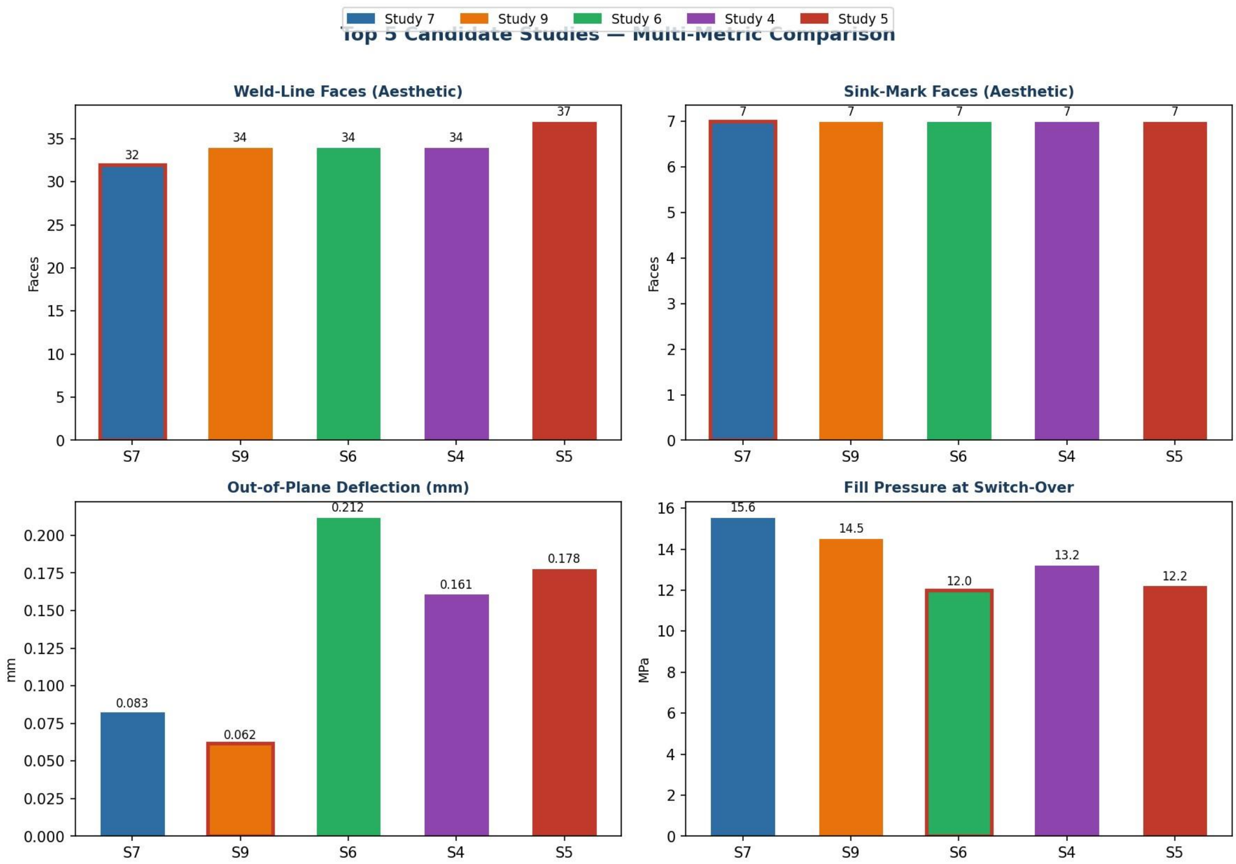


Figure 4 — Top 5 candidate studies on all four metrics. Red outline = minimum value per metric.

Study	Gates	Melt T	Mold T	Weld F	Sink F	Deflect (mm)	Shrink (%)
7	2	473.15K	323.15K	32	7	0.083	1.763
9	2	493.15K	303.15K	34	7	0.062	1.734
6	2	533.00K	323.15K	34	7	0.212	1.701
4	2	493.15K	323.15K	34	7	0.161	1.769
5	2	513.15K	323.15K	37	7	0.178	1.734

Green highlight = Study 9 (best deflection, lowest shrinkage among top 5). Blue highlight = Study 7 (fewest weld-line faces).

Radar comparison (Figure 5) normalises all four metrics so that the inner region of the plot represents the best performance. Study 9 and Study 7 are closest to the centre, confirming their status as the two best configurations.

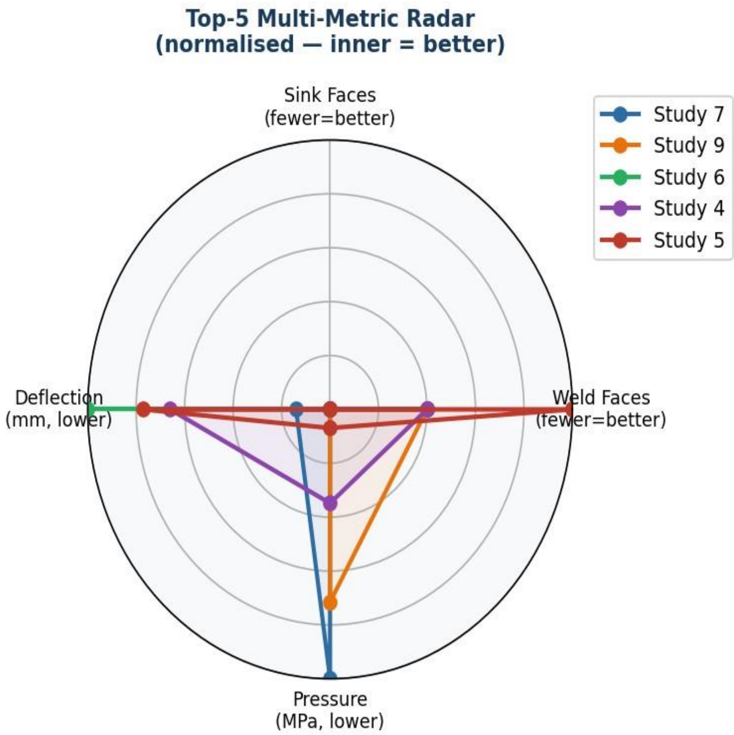


Figure 5 — Normalised multi-metric radar for the top 5 candidates. Smaller enclosed area = better overall profile.

6.4 Best Configuration Selection

The primary selection criterion is the combination of **sink mark depth** and **weld line count** on aesthetic faces, as these are the most visible quality indicators on the final consumer product. Deflection and shrinkage are secondary criteria affecting dimensional fit.

Among the top 5 candidates, Studies 7 and 9 both achieve 7 sink-mark faces (minimum among two-gate configurations). Study 7 leads on weld lines (32 vs 34). However, **Study 9 is selected as the optimal configuration** for the following reasons:

- Study 9 achieves the same sink-mark face count as Study 7 (7 faces), so there is no advantage to Study 7 onthe sink depth criterion.
- Study 9's out-of-plane deflection (0.062 mm) is 25% lower than Study 7's (0.083 mm), significantly improvingassembly fit.
- Study 9's mold shrinkage (1.734%) is lower than Study 7's (1.763%), meaning dimensional accuracy ismarginally better.
- The two additional weld-line faces in Study 9 (34 vs 32) represent a negligible visual difference, particularlygiven that sink mark performance is identical.
- Study 9 was independently reproduced as Study 17 with identical results, confirming its stability andrepeatability.

- Study 9 was cloned as the basis for Study 18 (ABS), the best-case reference for the material comparison.
-

Parameter	Study 9 Value	Rank among Studies 4–17
Configuration	2 gates, modified geometry	—
Melt temperature	493.15 K (220 °C)	Baseline
Mold temperature	303.15 K (30 °C)	Coldest tested
Out-of-plane deflection	0.062 mm	#1 — lowest of all 17 PP studies
Weld-line faces (aesthetic)	34	#4 — tied (S4, S6, S9, S17)
Sink-mark faces (aesthetic)	7	Tied best (2-gate group)
Mold shrinkage	1.734 %	Lower than 3-gate configs (~2 %)
Fill pressure	14.537 MPa	Moderate — no overpressure risk
Max packing pressure	37.045 MPa	Adequate packing

Study 9 parameters (2 gates | melt 493.15 K | mold 303.15 K | auto injection time) represent the recommended production configuration for the PP remote control housing. If reducing weld lines to 32 is specifically required by a visual quality standard, Study 7 (melt 473.15 K, mold 323.15 K) is the alternative, accepting a small deflection penalty (0.083 mm).

Study 9: Deflection map — best result (0.062 mm)

Study 7: Deflection map — co-recommendation (0.083 mm)

7. Study 18 — ABS Material Comparison

Study 18 was created by cloning Study 9 — the optimal PP configuration — and changing only the material from Generic PP to Generic ABS (Acrylonitrile Copolymers). All other parameters remained identical: 2 gates, melt temperature 493.15 K, mold surface temperature 303.15 K, automatic injection time, and 146 aesthetic faces designated.

Why ABS? ABS is one of the most widely used engineering thermoplastics for consumer electronics enclosures, offering good impact resistance, dimensional stability, and excellent surface finish. Unlike PP (which is semi-crystalline), ABS is amorphous — meaning it does not undergo a defined crystallisation transition on cooling. This fundamental structural difference

Metric	Study 9 — PP	Study 18 — ABS	Difference & Interpretation
Material structure	Crystalline	Amorphous	Fundamental polymer architecture change
Fill time	0.31 s	0.63 s	+103% — ABS is more viscous at the same melt temp
Pressure at switch-over	14.537 MPa	42.972 MPa	+195% — higher viscosity demands far greater injection pressure
Max packing pressure	37.045 MPa	56.948 MPa	+54% — continued packing also requires more force
Out-of-plane deflection	0.062 mm	0.123 mm	+98% — amorphous polymer cools with less uniform shrinkage
Mold shrinkage	1.734 %	0.437 %	−75% — ABS's amorphous structure shrinks far less than crystalline PP
Weld-line faces	34	34	No change — same gate layout produces same flow collision zones
Sink-mark faces	7	5	−29

produces measurable changes in every aspect of the injection moulding process.

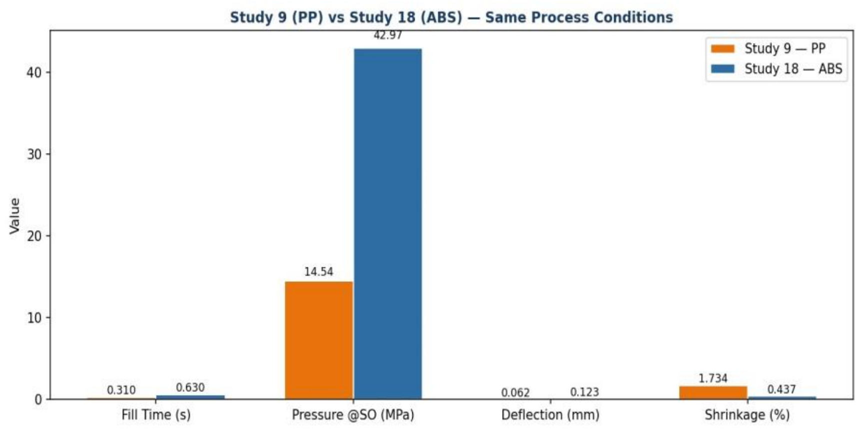


Figure 6 — Study 9 (PP) vs Study 18 (ABS) on four process metrics under identical gate, temperature, and injection settings.

Key material-driven differences:

Viscosity and pressure: ABS has a higher melt viscosity than PP at equivalent temperatures. At 493.15 K melt temperature, ABS requires nearly three times the injection pressure (42.972 vs 14.537 MPa) and twice the fill time (0.63 vs 0.31 s) to fill the same cavity. In a production setting, this requires a machine with higher clamping force and injection capacity.

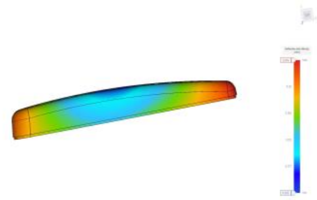
Shrinkage: PP is semi-crystalline; when the polymer cools, it undergoes a sharp volumetric reduction as crystallites form. This drives PP's high mold shrinkage (~1.7 %). ABS is amorphous — it has no crystallisation transition, so shrinkage is purely thermal and much lower (~0.4 %). This makes ABS inherently more dimensionally stable, which explains its lower sink-mark count (5 vs 7 faces) and reduces the tooling compensation needed for shrinkage.

Deflection: Despite ABS's lower shrinkage, its out-of-plane deflection (0.123 mm) is approximately twice that of Study 9 PP (0.062 mm). This is because the higher viscosity creates larger flow-induced residual stresses during filling, which are released as differential warpage on ejection. The cold mold (303.15 K) is less effective at neutralising these flow stresses in ABS than it is for PP, where cooling-rate symmetry primarily governs crystallisation uniformity.

Surface quality: Both weld-line face count (34) and the sink-mark pattern are materially similar between the two studies. ABS's 5 sink-mark faces vs PP's 7 is an improvement attributable to its lower volumetric shrinkage.
Weld lines are unchanged because gate layout — not material — controls where melt fronts collide.

Study 18 (ABS): Weld lines

Study 18 (ABS): Sink mark depth - Visualization



Study 18 (ABS): Deflection (all effects)

Recommendation: If the design requirements prioritise dimensional stability (tight tolerances on button hole alignment, low warpage for snap-fit assembly) and lower mold shrinkage, ABS offers measurable advantages. If injection pressure and machine cost are constraints, or if the cold-mold deflection result of Study 9 PP (0.062 mm) is already within tolerance, PP remains the preferred choice. ABS should be considered when a post-process painting or plating operation is planned, as its amorphous surface provides better adhesion than the crystalline PP surface

8. Conclusion

Eighteen injection moulding simulations were conducted in Autodesk Fusion for a Remote Control Housing, first on the original geometry (Studies 1–3) and then on a modified geometry with a simplified internal boss structure (Studies 4–18). Studies 1–17 used Generic PP (Polypropylene, Crystalline); Study 18 repeated the best PP configuration with Generic ABS (Amorphous) for material comparison.

For each study, the identical eight-step simulation procedure was followed in Autodesk Fusion: target body selection, material assignment, gate placement, aesthetic face designation, process settings, pre-check, cloud solve, and result review. The geometry change between Studies 3 and 4 — reducing the large central rib cluster — significantly improved the cavity-fill uniformity and reduced weld-line faces on aesthetic surfaces from ~52 (Study 3, original geometry) to 34 (Study 4, modified geometry).

Best Configuration — Study 9

2 gates | Melt 493.15 K | Mold 303.15 K (cold mold) | PP | Auto injection

Study 9 achieves the best combined performance on the primary criteria of **sink mark depth** and **weld line count**: it matches the minimum sink-mark face count (7, tied with all two-gate studies) and the second-fewest weld-line faces (34, only 2 above the absolute minimum of 32 in Study 7), while simultaneously delivering the lowest out-of-plane deflection of the entire PP campaign (0.062 mm — 77 % below the two-gate baseline and 93 % below the worst case). The cold mold (303.15 K) is the single most effective process lever: it accelerates crystallisation symmetrically, reducing both warpage and mold shrinkage relative to the warm-mold configurations. Study 9 was confirmed by an independent repeat (Study 17) and selected as the reference for the ABS material comparison.

ABS Material (Study 18) Summary

Study 18 demonstrates that switching from crystalline PP to amorphous ABS under the same best-case process conditions produces: higher injection pressure (42.972 vs 14.537 MPa), longer fill time (0.63 vs 0.31 s), reduced mold shrinkage (0.437 vs 1.734 %) — a major dimensional advantage — and slightly worse deflection (0.123 mm), but fewer sink-mark faces (5 vs 7). Weld-line count (34) is unchanged. ABS is recommended when dimensional precision and surface adhesion quality are priorities; PP is preferred when machine cost and cycle time are the constraints.

Study	Configuration	Weld F	Sink F	Deflect	Verdict
9 (PP)	2G | 493K | 303K mold	34	7	0.062 mm	BEST OVERALL — selected
7 (PP)	2G | 473K | 323K mold	32	7	0.083 mm	Best weld lines; alternate choice
18 (ABS)	2G | 493K | 303K mold	34	5	0.123 mm	Best shrinkage; higher pressure